



Real-Time Smart Room Monitoring & Data Pipeline

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Data Stream Collection and Simulation

PRIMARY DATASET

UCI Occupancy Detection: 20,560 sensor readings from a single office room.



RECORDED FEATURES & SPAN



Temp, Humidity, Light, CO₂, and Humidity Ratio recorded every ~1 minute.



Data spans Feb 4–18, 2015, split into training (8,143 records) and two test sets.

STREAM SIMULATION

Replaying records one-by-one in timestamp order.



Streaming K-Means

INITIALIZATION & LEARNING



One-pass online learning: Each record updates the model immediately with no batch reprocessing or storage



Cold-start initialization: First 2 records seed the initial centroids directly

MODEL UPDATES



Incremental centroid update: Running average formula ($\text{centroid} += (\text{x} - \text{centroid}) / \text{count}$)



Feature normalization: All 5 sensor features scaled by fixed domain maximums before distance computation

EVALUATION & LABELING



Majority-vote cluster labeling: Each cluster tracks true occupancy counts seen; dominant label is assigned at evaluation time to map unsupervised clusters to occupied/unoccupied



Accuracy reported: Every 30 records during the stream and as a final score at completion

Next steps



Implement DBSCAN — proposal planned to explore density-based clustering as an alternative to K-Means for handling irregular room-state boundaries



Add silhouette score — evaluate internal cluster quality and separability beyond just accuracy



Expand evaluation metrics — compute precision, recall, and F1-score via post-hoc labeling to get a fuller picture of model performance



Anomaly/outlier flagging — detect unusual sensor readings that fall outside normal occupancy patterns